General Relativity:

A Programmer's Guide to Curved Spacetime

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▮ Equivalence Principle and Spacetime Curvature ▮

Legend says, one day Einstein, while observing a window washer on a ladder near his office in Bern, Switzerland started to imagine what would happen if the worker was to fall. He didn't imagine what would happen to the washer when he touches the ground, but what he would experience as he would be falling. Einstein realized that if the worker was to fall, the gravity ($\boldsymbol{9.8m/s^2}$) would have been the only force acting on him for him (in his/worker's respective perspective). The washer wouldn't *feel* the gravity, he'll just experience it. And since the washer is falling, there will be no contact forces, hence the washer would have felt completely "weightless", as if he is floating in space! This is known as the "Principle of Equivalence".

This made Einstein more curious, and he started to ask himself if there was a way to tell the difference between the gravity on Earth and the acceleration in space. Specifically, this thought aroused after he imagined being in a room on Earth and checking his weight using a scale and being in a room on a rocket in the space and checking his weight in a scale, which is accelerating exactly at the speed of gravitational acceleration on Earth. The weight at both scenarios will be the exact same!.

After he questioned himself, he imagined what would happen if he pointed a flashlight from one end of the room to another, as the rocket was accelerating upwards. Soon he noticed that the height of the light at the other side of the room, would be slightly lower than from the source of the light. Since the room was accelerating upwards by $\boldsymbol{9.8m/s^2}$, the beam of light appeared to be curved downwards at the other end of the room. However, same flashlight experiment done on Earth would have resulted in the same height at both ends of the flashlight. It didn't made sense to Einstein at the time, because it violates the Principle of Equivalence.

Then to save the principle of equivalence, Einstein realized that the light "MUST" curve in gravitational field as well, which according to Newton wasn't possible, because Newton said that "The gravity is a force that pulls on objects with *Mass*!" And light, having no mass seemingly can't bend in gravity, right? Well, for the principle equivalent to be saved, Light must bend but It can't. So, Einstein's answer was that the gravity must be bending space itself! This meant that gravity is *not* a force pulling on the objects, instead massive objects (for example a sun) warps the geometry of space. This showed that the light will follow the curved space, and hence the principle of equivalence is saved!

Now, to express all this theory mathematically, it required complex geometrical mathematics, for which Einstein shook his hands with an old buddy of his, a Mathematician, named Marcel Grossman. And with his help of his PhD in the Geometry of Curved Spaces (known as Reimannian geometry), Einstein got the mathematical understanding of how curved spaces work. Then, after approximately 2-3 years Einstein came up with this:

$$G_{\mu\nu} + \Lambda \cdot g_{\mu\nu} = \frac{8\pi G}{c^4} \cdot T_{\mu\nu}$$

where:

01 - $\boldsymbol{G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} \cdot R \cdot g_{\mu\nu}}$

where:

001 - $\boldsymbol{R_{\mu\nu}}$: A $\mathbf{4 \times 4}$ Matrix that describes how space-time is stretched. It is known as the "Ricci Curvature Tensor".

002 - $\boldsymbol{-\frac{1}{2} R \cdot g_{\mu\nu}}$: Half of the " Scalar Multiplication" of the scalar $\boldsymbol{R}$ (The total curvature at a point) and a metric tensor ($\boldsymbol{g_{\mu\nu}}$) which is also a $\mathbf{4 \times 4}$ Matrix, that tells us how far two nearby events are. Both of these **001** and **002**, results in an "Einstein Tensor ($\boldsymbol{G_{\mu\nu}}$)".

02 - $\boldsymbol{\Lambda \cdot g_{\mu\nu}}$: The Scalar Multiplication of a "Cosmological Constant" that represents vacuum energy density (known as Dark Energy) and the same metric tensor.

03 - $\frac{8\pi G}{c^4}$: Multiplication of a number $\boldsymbol{8\pi}(\approx \mathbf{25.13})$ with Newton's Gravitation Constant '$\boldsymbol{G}$' $(\mathbf{6.674 \times 10^{-11}} \boldsymbol{m^3/(kg \cdot s^2)})$, divided by the speed of light to the fourth power $(\mathbf{8.1 \times 10^{33}} \boldsymbol{m/s^4})$. This fraction shows how much spacetime curves per unit of mass/energy.

04 - $\boldsymbol{T_{\mu\nu}}$: A stress energy $\mathbf{4 \times 4}$ Matrix that describes all the matter and energy in spacetime.

(NOTE: THIS IS JUST A GLIMPSE TO THE EINSTEIN FIELD EQUATION, IT WON'T BE UNDERSTANDABLE YET, BUT WILL BE EXPLAINED LATER IN THE PDF.)

Another important note here is that the trampoline illustrations (2d plane) shown for the spacetime is for visualization only, in reality the actual interaction of gravity and matter is in 3 dimensions.

Before Einstein published this theory in 1915, he actually published a preliminary calculation, predicting that light will bend near the Sun. His calculation was same as the Newton's theory (**0.87** Arc seconds), in this preliminary calculation released in 1911. After that, in 1915, he re-did the calculation and it turned out to be exact double of 1911's, **1.75** Arc seconds!

After that came a challenge, Einstein published his theory in 1915, which was being ignored till 1918, as The World War I was almost ending. On May 29 1919, Two teams were going for an expedition to observe the total solar eclipse that day. Team 1: Sobral, Brazil was led by Andrew Crommelin and Charles Davidson. Team 2: Principal Island, West Africa (Gulf of Guinea) was led by Arthur Eddington (Perfect as he had the knowledge of the general relativity) and Edwin Cottingham.

There were two teams because if one finds bad weather the other might succeed. These two teams actually came to their destination months before, and on May 29, 1919, it turned out to be too cloudy for Eddington's team. The moon covered the Sun Totality at 2:13 PM, and then at 2:18 PM the totality ended! They only got 16 pictures in these five minutes, from which only two weren't cloudy and useless. But, for the Crommelin's team, they found the perfect weather! They took multiple perfect photographs.

Then near June 1919, Eddington compared the stars position to the reference photographs which were taken when the Sum wasn't present. Eddington did some calculations and got to know that the stars had shifted approximately **1.6** Arc seconds! Which was very near to the Einstein's measurements. This was announced on a Joint meeting laid on November 6, 1919, after which Einstein was not only accepted, but also famous.

▮ Special Relativity: The Revolution ▮

Firstly let us know what is the principle of relativity. Imagine a bench where few people are sitting and staring at the ongoing trains. The train is for example moving in the forward Direction and in a straight line with a constant speed. For the people in the train, the train will look (not feel, look) motionless and they will see other things related to them moving behind (opposite to the direction of the train). Whereas for the people at the bench, they will look at the train moving and they're motionless. Remember as long as the motion is of straight line with a constant speed, it is acceptable to nod yes to either of these perspectives. This was the first postulate of special relativity.

Special Relativity, actually, is a physical theory that describes how space and time work. For example, if we as a train passenger, throw a ball forward while in the train, for us the ball is traveling at a constant speed, but for the people on the bench, outside the train, the ball is traveling forward faster, as the impulsion of the ball and the moving train combines. However, doing same experiment with light through for example a flashlight instead of a ball, we notice that the light will travel in the same speed for both perspectives, which gives ammunition to the fact that the speed of light is a universal constant value. This light example was the second postulate of special relativity.

From basic physics, we already know that speed ($\boldsymbol{v}$) multiplied by time($\boldsymbol{t}$) gives us distance ($\boldsymbol{d}$). But, when it is about light we know that it is a universal constant. Hence, distance and time will need to change, but why? Think of it like this: imagine a spaceship going at $\boldsymbol{0.9c}$ (c is the speed of light) or 90 percent of the light speed. If someone in the spaceship shines a flashlight forward, the classic physics will say that the flashlight's light will travel at: $\boldsymbol{0.9c + c = 1.9c}$ due to velocity addition. But that's not right! The the flashlight will just travel at c! Hence to preserve the constancy of the speed of light, space and time must adjust. And, when these distances adjust themselves, its a "Length Contraction", and when the time adjusts itself it is "Time Dilation".

First, let us try to understand the Time Dilation. Let's say we have two 'Light clocks', which consists of glass at the top and the bottom, with a ball bouncing between these glasses. When the ball returns to a glass after already touching it before, it is a 'Tick' on the clock. Let's say one of those light clock is stationary, while the other is in a moving body. The light clock which is stationary will have a normal tick speed, whereas for the light clock in a moving body such as a train, the ball will travel diagonally forward or in the direction of the movement, hence more distance, hence more time required for a tick, hence the tick speed will be slower.

(NOTE: LIGHT IS A UNIVERSAL CONSTANT, TIME DILATION OCCURS DUE TO MORE DISTANCE REQUIRED TO TRAVEL, WHICH INCREASES THE TIME REQUIRED FOR AN EVENT TO TAKE PLACE.)

Now we have Length Contraction. Length contraction is when something moving relative to you will seem shorter to the direction it is moving to. Let's take an example: If we measured a train to be $100m$ long, when the train wasn't moving. It will look more squished to the direction of it's movement, when it moves.

▮ Tensors ▮

In the past chapters, we learned about several terms like, a '*Tensor*', or a '*Matrix*' or a '*Scalar*'. All these come under a Tensor. A Tensor is a multi-dimensional array/grid of numbers, it is a mathematical object that can be a generalization of 'Scalars', 'Vectors', and 'Matrices' (plural of a Matrix). Each of these dimensional arrays have a Rank, denoted by $\boldsymbol{R}$. Let dig in to each of these

A Scalar or a 'Rank-0 Tensor' is nothing more than a number, like $\boldsymbol{\pi}$, 1, or even a fraction like $\boldsymbol{\frac{3}{11}}$. We generally don't call it an array as it is just a single number. A scalar is denoted as $\boldsymbol{S}$.

A Vector or a 'Rank-1 Tensor' is a one-dimensional array, a list of numbers extending downwards. This can be shown like this:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \cdots v_n \end{bmatrix} \updownarrow \textit{One-Dimensional}$$

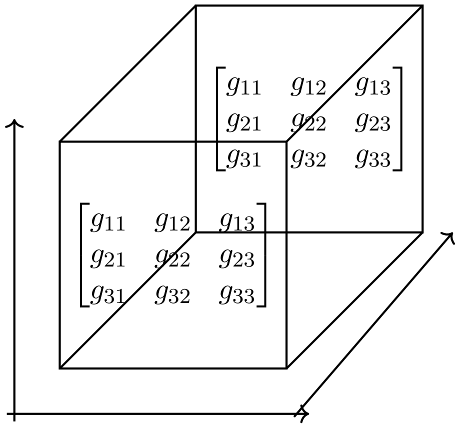
A Matrix or a 'Rank-2 Tensor' is a two-dimensional array, a list extending both towards right and down. This can be shown like this:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix}$$

Two-Dimensional

Matrix

A 'Rank-3 Tensor' is a three-dimensional array of numbers, and when we speak of Tensors, we generally mean this Rank-3 tensor, or in mathematical terms, we generally mean an array of more than two dimensions ($\boldsymbol{R > 2}$). This can be illustrated like:



Three-Dimensional

These illustrations were just a few, Tensors are n-dimensional arrays. These illustrations were just the array definition of Tensors, these tensors also have a meaning behind them, a geometrical meaning.

▮ Vectors, Co-vectors, and Basis ▮

This is the starting of the mathematics behind the General Relativity. Starting with a simple vector; A Vector, as we have studied in basic Physics, is a quantity that has both direction and *magnitude*. The notation of a Vector looks like this: $\boldsymbol{\vec{V}}$. The arrow at the top is nothing but a factor of distinguishability of a Vector.

A Basis is a set of *reference* Vectors, we use to describe other Vectors. For example, $\boldsymbol{\vec{e_1}}$ and $\boldsymbol{\vec{e_2}}$; are two Vectors between which: $\boldsymbol{\vec{e_1}}$ is a Vector along y-axis, and $\boldsymbol{\vec{e_2}}$ is a Vector along the x-axis. These are called the 'Basis Vectors'. Let us know understand what an equation like this would mean:

$$\vec{V} = 3\vec{e_1} + 5\vec{e_2}$$

These numbers: 3 and 5 are called 'Components'. Components are just numbers, Scalars. To get an actual geometrical arrow (Vector) as a result, we multiply the components with the basis vectors. The equation above says: "Go three steps in the y-axis + Go five steps in x-axis".

A more compact form of this Vector with Component notation, will be:.

$$\vec{V} = V^1 \times \vec{e_1} + V^2 \times \vec{e_2}$$

where:

$\vec{V}$ = The final 'Displacement Vector'.
V^1 = The component of $\vec{e_1}$.
$\vec{e_1}$ = A Basis Vector.
V^2 = The component of $\vec{e_2}$.
$\vec{e_2}$ = A Basis Vector.

Now's the time for a big example:

- Imagine a drone flying in a 2D space. The Basis Vectors we take are: $\boldsymbol{e_1}$ and $\boldsymbol{e_2}$ for East (Right) and North (Up) directions respectively.

- You see the drone moving 6 meters towards East and 8 meters towards North. Hence, the components for our Basis Vectors will be:

- The displacement Vector $\boldsymbol{V}$ will be:

$$\vec{V} = V^1 \times \vec{e_1} + V^2 \times \vec{e_2}$$

  Substitute the values:

$$\vec{V} = 6\vec{e_1} + 8\vec{e_2}$$

- Now, the drone rotated 90° clockwise for you friend's perspective. The basis will change now: $\boldsymbol{\vec{e_1}'}$ and $\boldsymbol{\vec{e_2}'}$, which were pointing at East (Right) and North(Up) before, respectively, but will now point at North(Up) and West(Left) after the rotation, respectively. The symbol $\boldsymbol{'}$ is a 'Prime Symbol', that indicates a difference in Basis or 'Coordinate System'.

A Co-vector on the other hand (also known as a dual-vector or a One-form) is a linear function that asks for Vectors as 'input' and outputs a Scalar. It's not an arrow pointing at a direction, it's a function. For example, a grocery list saying a '\$3 per apple and \$5 per orange' has the co-vectors $\mathbf{(3, 5)}$. Co-vectors are, in simple words a set of rates. The notation of co-vectors is this: $\boldsymbol{\omega_1}$, this W looking symbol's name is 'Omega'. So a vector equation like this:

$$\vec{V} = V^1\vec{e_1} + V^2\vec{e_2}$$

will look like this, in terms of co-vectors:

$$\tilde{\omega} = \omega_1\tilde{e}^1 + \omega_2\tilde{e}^2$$

where $\boldsymbol{\tilde{\omega}}$ is the actual dual vector, and $\boldsymbol{\omega_1}$ and $\boldsymbol{\omega_2}$ are the components of the 'Dual Basis Vectors'. Dual Basis Vectors are basis vectors that we use for the co-vectors. To differentiate the Vectors basis from co-vectors basis (in terms of notation), we use a 'tilde' symbol over the vector, instead of an arrow.

▮ Vector and Co-vector's Transformations ▮

Transformation means the change in coordinate system or the basis. Think of it like switching from Meters (m) unit to Feet (ft) unit. Transformation changes the component's values. Each, vector or a co-vector, has it's unique transformation:

▮ Vector Transformations ▮

Imagine having a stick of two meters long. The basis Vector we will make will be: $\boldsymbol{\vec{e} = 1}$ (a basis Vector with the length of 1 meter). Thus, the component will be: $\boldsymbol{v^1 = 2}$ .The final stick equation now will be:

$$\vec{V} = 2\vec{e_1}$$

But, now we change the unit from meters to centimeters. Hence, the unit of basis Vector also went to cm from m. Therefore, we can say that our basis Vector got smaller. Now, because our basis Vector got smaller, our component has to be 200 centimeters, to make up for the two meters long stick. Thus, we can conclude that in vector transformation when basis vectors get smaller, the component gets bigger! We call this 'Contra-variant.'

▮ Co-Vector Transformations ▮

Imagine you are upon a hill, temperature there increases as you go West. The gradient will be, let's say every meter on East the temperature increases by 5 degrees Celsius. We can write it like: $\boldsymbol{\tilde{e}^1} =$ **"**How much West per Meter**"**. We can make our component: $\boldsymbol{\omega_1 = 5°/m}$. But if we shift from meters to centimeters, we can see that basis vector and component both decreasing! We call this 'Covariant'

▮ Vector and Co-vector's Spaces ▮

Vectors, as we have already learned, are elements of a vector space ($\boldsymbol{V}*$), where they live. $\boldsymbol{V}$ is the collection of objects that we call 'Vectors', along with the rules on how to combine them. The key requirements are that, you can add any two vectors together and get another Vector in that space, and you can multiply any vector by a scalar to get another Vector in that space.

And Co-vectors are elements of a dual space or a Vector star space ($\boldsymbol{V*}$). $\boldsymbol{V}*$ is where co-vectors live. $\boldsymbol{V*}$ is the collection of all possible 'linear functionals' that can act on vectors from $\boldsymbol{V}$.

An element from $\boldsymbol{V}$ is something like $V = (x, y, z)$, depending on $\mathbb{R}$, which is the set of real numbers. If $\boldsymbol{V} = \mathbb{R}^{\boldsymbol{3}}$, then $\boldsymbol{V}$ can be like $\boldsymbol{(1, 2, 3)}$, $\boldsymbol{(5, 9, 12}$, and so on.

An element from $\boldsymbol{V*}$ is like $\boldsymbol{\omega}$ where $\boldsymbol{\omega(V) = 2x + 3y + z}$, and so on. We call them dual, because we can combine them: Co-vector(Vector) = Scalar. This might not be clear yet, as we didn't discuss what 'Linear Functionals' are.

A Linear Functional is a special type of function that takes a Vector as an input, and gives us a single Scalar as output, in a 'linear' way. Linear way means it obeys the following rules:

▮ **01.** $\boldsymbol{f(c + w) = f(v) + f(w)}$ ▮

The Function of sum is equal to the sum of functions

▮ **02.** $\boldsymbol{f(c \cdot v) = c \cdot f(v)}$ ▮

Stretching the input by some factor, stretches the output by the same factor. For example:

$$V = (1, 2, 3)$$
$$f(x, y, z) = 2x + 3y + z$$

Now we plug in the Vector:

$$f(V) = 2(1) + 3(2) + (3)$$
$$=> 11$$

And, now if we Double our Vector from $(1, 2, 3)$ to $(2, 4, 6)$:

$$f(2V) = 2(2) + 3(4) + (6)$$
$$=> 22$$

That's a perfect double in Scalar as well!

Hence, these functions that are linear, lives in **V***.

We know that Vectors live in $\boldsymbol{V}$ space. and Co-vectors live in $\boldsymbol{V*}$ space. How do we now take two Vectors and multiply them to get a scalar? We use Inner Products!.

Inner Product is a fancy way of saying "Multiply two vectors and get a scalar". We notate this function like this $\langle \boldsymbol{v}, \boldsymbol{w} \rangle$. Inner products are just generalization of Dot products. For example:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \cdot \quad \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$
$$= (1 \times 4) + (2 \times 5) + (3 \times 6) = 32!$$

Looking at $\boldsymbol{= (1 \times 4) + (2 \times 5) + (3 \times 6)}$, we can say that is exactly same as a Linear Functional: $\boldsymbol{f(w) = (1 \times w_1) + (2 \times w_2) + (3 \times w_3)}$. Since $\boldsymbol{w = (4, 5, 6)}$, we, if plug in $\boldsymbol{w}$ to out Linear Functional, will get the same scalar 32!. So basically, Inner Products are secretly used to turn our $\boldsymbol{V}$ vector into a Co-vector.

Now, let us explore 'Metric Tensors'. Metric Tensors are mathematical tools that tell us how to measure distances and angles in any space, especially curved ones. In a simple flat space like a grid, we can use pythagorean theorem to find the distances. But, on a curved space like a Globe, that rule can't work. The metric tensor provides the 'Correctional Factors' at every single point. Hence, the pythagorean theorem works in the curved examples as well. Metric Tensor, in simple words, is a table of correction factors that depend on the shape of the grid. We call this 'table' $\boldsymbol{g}$. We multiply this table's elements with the sum of all possible combination of opposite and adjacent sides.

We have a difficult notation for these Metric Tensors using indices, a $\boldsymbol{g_{\mu v}}$, which we took a glimpse of in Einstein's Field equation. Metric Tensors are rank-2, hence a Matrix, and they look something like this:

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$$

The index/indice $\boldsymbol{\mu}$ tells us the row of the tensor and the second indice $\boldsymbol{v}$ tells us the column. Hence, $\boldsymbol{g_{11}}$ would be row 1 and column 1

Let's take an example of a 2D planes:.

- $\boldsymbol{x^1}$ tells us how far we traveled in direction 1 (Horizontal, x-axis)

- $\boldsymbol{x^2}$ tells how far we traveled in direction 2 (Vertical, y-axis)

- Let's say our metric tensor is:

$$\begin{array}{cc} & \begin{array}{cc} \nu=1 & \nu=2 \end{array} \\ \begin{array}{c} \mu=1 \\ \mu=2 \end{array} & \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \end{array}$$

$$\begin{aligned} \boldsymbol{dx^1} &= \text{tiny move in direction 1} \\ \boldsymbol{dx^2} &= \text{tiny move in direction 2} \end{aligned}$$

Hence $\boldsymbol{g_{11}} = 1$, $\boldsymbol{g_{12}} = 0$, $\boldsymbol{g_{21}} = 0$, $\boldsymbol{g_{22}} = 1$

- Let's see what these values mean in a physical world:

01 - $\boldsymbol{g_{11}}$: How much moving in direction 1 contributes to the entire distance?.

02 - $\boldsymbol{g_{12}}$, $\boldsymbol{g_{22}}$: Used to describe the 'slant' or angle between measuring the axes (x and y axes are perpendicular in 2d grid space, so no problem there). These two are the same always.

03 - $\boldsymbol{g_{12}}$, $\boldsymbol{g_{21}}$: Same as $\boldsymbol{g_{11}}$, is like a component, we multiply it's value to our terms involving in the second direction squared, only.

Now, we will talk about the equation of "Spacetime Line Element". An Equation used to calculate the infinitesimal(very small) distances between two points. The equation looks like this:

$$ds^2 = g_{\mu\nu} \cdot dx^\mu \cdot dx^\nu$$

where:

01 – $\boldsymbol{ds^2}$: Is the infinitesimal squared distance between two points.

02 – $\boldsymbol{g_{\mu\nu}}$: Is the Metric Tensor.

03 – $\boldsymbol{dx^\mu}, \boldsymbol{dx^\nu}$: Are the infinitesimal coordinate's differential changes in the position, of the same coordinate system.

Let us take the example of a "Polar coordinate System". Polar coordinates locates a point using a distance ($\boldsymbol{r}$) from a a central point ('pole'), with an angle ($\boldsymbol{\theta}$). This is an alternative to the "Cartesian Coordinate System", which uses x and y-axes which we all gave learned in Basic Mathematics. The Metric Tensor for flat 2D space in Polar Coordinates is:

$$\boldsymbol{g_{\mu\nu}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

where:

01 – **1**: Is $\boldsymbol{g_{rr}}$, distance from the central pole.

02 – **0**: Is $\boldsymbol{g_{r\theta}}$ and $\boldsymbol{g_{\theta r}}$, ignored as coordinate system is orthogonal.

03 – $\boldsymbol{r^2}$: $\boldsymbol{dx^\nu}$: Is $\boldsymbol{g_{\theta\theta}}$, the angle

Using the spacetime line element, we can substitute all the metric tensor's values in, like this:
[Note: $\boldsymbol{dx^r}$ means $\boldsymbol{dr}$, and $\boldsymbol{dr^\theta}$ means $\boldsymbol{d\theta}$]

$$ds^2 = g_{\mu v} \cdot dx^\mu \cdot dx^v$$

$$\implies ds^2 = g_{rr} \cdot dx^r \cdot dx^r + g_{r\theta} \cdot dx^r \cdot dx^\theta + g_{\theta r} \cdot dx^\theta \cdot dx^r + g_{\theta\theta} \cdot dx^\theta \cdot dx^\theta$$

$$\implies ds^2 = g_{rr}(dr)(dr) + g_{r\theta}(dr)(d\theta) + g_{\theta r}(d\theta)(dr) + g_{\theta\theta}(d\theta)(d\theta)$$

$$\implies ds^2 = 1(dr)^2 + \cancel{0(dr\theta) + 0(d\theta r)} + r^2(d\theta)^2$$

$$\implies \boxed{\mathrm{ds}^2 = dr^2 + r^2 d\theta^2}$$

▮ Minkowski Metric Tensor ▮

The Minkowski Metric Tensor is made for flat spacetime in special relativity. In 3d spaces, we can easily measure distances between two points. But, in spacetime (4D: 3d spacial + 1 time), we measure the 'Interval' between two 'Events' An Interval is just a normal distance for spacetime, and an Event is just something that happens at a specific place and specific time. The Minkowski Metric Tensor in is standard form is like this:

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$$

This formula is called the "Spacetime Interval" formula (Infinitesimal form). Which is mainly used for theory and Calculus. Let us now deep dive into each of these sets and understand them properly:

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$$

where:

01 – $\boldsymbol{ds^2}$: Is the Interval Squared.

02 – $\boldsymbol{-c^2dt^2}$: Is negative speed of light squared multiplied by the infinitesimal change in time (Time Differential). We use $\boldsymbol{c^2}$ to convert time into length units to compare it with other spacial distances. We use the negative sign to differentiate between ordinary space. It means Time and Space has opposite signs.

03 – $\boldsymbol{dx^2}$: Is spacial Differential in X-axis.

04 – $\boldsymbol{dy^2}$: Is spacial Differential in Y-axis.

05 – $\boldsymbol{dz^2}$: Is spacial Differential in Z-axis.

If we label $\boldsymbol{x^0 = ct}$, $\boldsymbol{x^1 = x}$, $\boldsymbol{x^2 = y}$, and $\boldsymbol{x^3 = z}$, the Metric tensor components will be:

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This is often called $\boldsymbol{\eta}$ (eta) rather than a $\boldsymbol{g}$ to distinguish it as a flat spacetime metric.

Once we have covered the $\boldsymbol{\eta}$, it's good to know that there are two sign conventions for Minkowski Metric Tensor:

01. $\boldsymbol{(-,+,+,+)}$ "Mostly Plus" : The one I showed, where Time is negative and Spacial Distances are Positive.

02. $\boldsymbol{(+,-,-,-)}$ "Mostly Minus" : Where Time is Positive and Spacial Distances are Negative.

Particle physicists prefer the first convention, whereas general relativists prefer the second convention. Either are fine and works properly, as they obey the rule of Time having a different sign then Spacial Distances.

Another formula to look gere is the 'Finite Version' of Minkowski Interval formula:

$$\Delta s^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

Both formulae are the exact same, this just look simpler with this ($\boldsymbol{\Delta}$) Delta sign. This formula will gave the exact same result as the older version. We will still use the Infinitesimal Form more, as the ($\boldsymbol{\Delta}$) Delta can't keep up if curvature is introduced.

Going deep, let us now discover the three types of spacetime Intervals:

▮ Timelike Intervals ($ds^2 < 0$) ▮

Let's say we have two events. Event1: You snap your fingers at $\boldsymbol{x = 0}$ and $\boldsymbol{t = 0s}$. Event2: You clap your hands at $\boldsymbol{x = 100}$ and $\boldsymbol{t = 1s}$. The $\boldsymbol{\Delta t}$ will be 1, $\boldsymbol{\Delta x}$ will be $100m$ and $\boldsymbol{\Delta y}$ & $\boldsymbol{\Delta z}$ remain zero (ignorable for this example). The time separation $\boldsymbol{c \cdot \Delta t}$ will be $\boldsymbol{3 \times 10^8 m}$, exactly $\boldsymbol{c}$! Whereas, the space separation $\boldsymbol{\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}}$ will be $100m$. Hence, the time separation or light is 'WAY MORE' than the spacial distance. Now using the finite Interval formula:

$$\Delta s^2 = -c^2(1)^2 + (100)^2 + (0)^2 + (0)^2$$
$$\implies \Delta s^2 = -(3x10^8)^2 + (100)^2 \approx -9x10^{16}m^2$$

We get a negative value! And Since the spacial distance (100m) is less than what light can travel ($\boldsymbol{c}$), we could easily walk from event 1 to event 2, send a light signal from event 1 to event 2. To sum it up, Event 1 CAN CAUSE Event 2. For example, a button press at $\boldsymbol{x = 0}$ and $\boldsymbol{t = 0}$ can make a rocket explode at $\boldsymbol{x = 100m}$ and $\boldsymbol{x = 1s}$

▮ Spacelike Intervals ($ds^2 > 0$) ▮

Again, Let's say we have two different events: Event1: A random star explodes around $\boldsymbol{x = 0}$ and $\boldsymbol{t = 0s}$. Event2: You sneeze on Earth at $\boldsymbol{x = 10^{16}m}$ away at $\boldsymbol{t = 1s}$. The $\boldsymbol{\Delta t}$ here will be 1 second, $\boldsymbol{\Delta x}$ will be $\boldsymbol{10^{16}m}$, and $\boldsymbol{\Delta y}$ & $\boldsymbol{\Delta z}$ will again be ignored as they are 0. The time separation $\boldsymbol{c \cdot \Delta t}$ here also will be $\boldsymbol{c \cdot (3 \times 10^6)m}$. And the space separation will be $\boldsymbol{\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2)} = 10^{16}m}$. Hence, the spacial distance is WAY MORE than light could travel: Now, again using the finite Interval formula:

$$\Delta s^2 = -c^2(1)^2 + (10^{16})^2 + (0)^2 + (0)^2$$
$$\implies \Delta s^2 = -(3x10^8)^2 + (10^{16})^2 \approx 10^{22}m^2$$

A positive value! And since the spatial distance is more than what light could travel, no signal from Event 1 can reach Event 2, or vice versa. To sum it up, Nothing can connect these two events together. Both events are completely independent.

!

▮ Lightlike Intervals ($ds^2 = 0$) - NULL ▮

For the last time now, Let's say we have two different events: Event1: Sun emits a quantum of light at $\boldsymbol{x = 0}$ and : $t = 0$. Event2: The same 'Photon' hits Earth at $\boldsymbol{x = 1.5 \times 10^{11} m}$ and $\boldsymbol{t = 500 s}$. The $\boldsymbol{\Delta t}$ here will be 500 seconds, $\boldsymbol{\Delta x}$ will be $\boldsymbol{1.5 \times 10^{11} m}$, and both $\boldsymbol{\Delta y}$ and $\boldsymbol{\Delta z}$ will be 0. The time separation ($\boldsymbol{c \cdot \Delta t}$) will be $\boldsymbol{1.5 \times 10^{11} m}$, and the space separation $\boldsymbol{\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2)}}$ will be $\boldsymbol{1.5 \times 10^{11} m}$! Hence, exactly the same! Let's see it with the same formula:

$$\Delta s^2 = -c^2(1.5 \times 10^{11})^2 + (1.5 \times 10^{11})^2 + (0)^2 + (0)^2$$
$$\implies \Delta s^2 = -(3 \times 10^8)^2 \cdot (500)^2 + (1.5 \times 10^{11})^2 = 0$$

This is the path of light! All photons travel along Null Intervals. Event 1 and Event 2 are connected by light traveling at exactly $\boldsymbol{c}$! This is the boundary between timelike and spacelike intervals.

▮ Lorentz Transformation ▮

Lorentz transformations are the equations that tell you how to convert space-time coordinates between inertial reference frames (moving relative to each other). Think of it like this for example:

Frame $\boldsymbol{S}$: You standing Still.

Frame $\boldsymbol{S'}$: George moving at Velocity $\boldsymbol{v}$ in $\boldsymbol{x}$.

The Pre-Einstein Galilean Transformations said:

- $\boldsymbol{x' = x - vt}$

- $\boldsymbol{y' = y}$

- $\boldsymbol{z' = z}$

- $\boldsymbol{t' = t}$

Lets break them down:

▮ Space Transformation: $\boldsymbol{x' = x - vt}$ ▮

$\boldsymbol{x}$ is the position of the event in YOUR frame. $\boldsymbol{v}$ is George's velocity, $\boldsymbol{t}$ is the time in your frame, hence $\boldsymbol{vt}$ is the distance George covers. Thus, this transformation is the position relative to George. For example, an event happened 100m away from your frame in 2 seconds, when Alice was going at 20m/s. So for her the position of the event will be 60m away, in her respective frame. which we know using

$$\boldsymbol{x - vt \implies (100m) - (20m/s)(2s) = 60m}.$$

▮ Perpendicular Transformations: $\boldsymbol{y' = y, z' = z}$ ▮

These perpendicular distances remain unchanged, because the motion in this example is on in x direction.

▮ Time Transformation: $\boldsymbol{t' = t}$ ▮

THIS IS WRONG! This assumes absolute time! But, this contradicts with the constancy of the speed of light.

Now that we know what was wrong in the Galilean transformation, we should first know the Lorentz factor, and how we calculate it. The formula is:

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

where:

$\boldsymbol{\frac{v^2}{c^2}}$: Is the Velocity Ratio Squared.

This Lorentz Factor ($\boldsymbol{\gamma}$) is used to measure the strength of 'Relativistic Effects'. Relativistic Effects are the strange physical phenomena that occur when objects move at speeds comparable to light speed. Now, let us see the new modified Lorentz transformation:

- $\boldsymbol{x' = \gamma(x - vt)}$

- $\boldsymbol{y' = y}$

- $\boldsymbol{z' = z}$

- $\boldsymbol{t' = \gamma(t - \frac{vx}{c^2})}$

Let's break these down as well:

▮ $\boldsymbol{x' = \gamma(x - vt)}$ ▮

This is quite similar to the classical Galilean x transformation. We just multiply that all by the Lorentz Factor. The reason we multiply $\boldsymbol{x - vt}$ by $\boldsymbol{\gamma}$ is because we need to respect the Einstein's Second postulate, that the light must be $\boldsymbol{c}$ for everyone.

▮ $\boldsymbol{y' = y, z' = z}$ ▮

These aren't even changed, same as it was in Galilean transformations.

▮ $\boldsymbol{t' = \gamma(t - \frac{vx}{c^2})}$ ▮

This is the main Einstein's Revolution: Time depends on position! We subtract the time on our frame by the 'Position-Dependent Correction' ($\boldsymbol{\frac{vx}{c^2}}$). This single fraction has the unit of time (seconds). The Position-Dependent Correction means that the time George measures depends on where the EVENT happened (x coordinate). The farther the event (larger x), the larger the correction ($\boldsymbol{\frac{vx}{c^2}}$). This creates the 'Relativity Of Simultaneity'. The physical meaning of this is that two events at the same time ($\boldsymbol{t}$), but different position ($\boldsymbol{x_1, x_2}$) in our frame, will have different times ($\boldsymbol{t'_1, t'_2}$) in George's frame. The effects of position are:

01 - $\boldsymbol{x > 0}$ - Event Ahead.

02 - $\boldsymbol{x < 0}$ - Event Behind

03 - $\boldsymbol{\frac{vx}{c^2} < 0}$ - Later time, after,
- will happen -

04 - $\boldsymbol{\frac{vx}{c^2} > 0}$ - Earlier time, before,
- will happen -

A Four-Vector is a Mathematical Object with four components that transform in a specific way under Lorentz transformations. One of these component is of time, while other 3 are spatial. The general form of a Four-Vector is like this:

$$A^{\mu} = (A^0, A^1, A^2, A^3)$$

Where $\boldsymbol{\mu = 0, 1, 2, 3}$. Here, $\boldsymbol{A^0}$ is the time-like component (often scaled by the speed of light $\boldsymbol{c}$, to have the unit of length), $\boldsymbol{A^1}$, $\boldsymbol{A^1}$ and $\boldsymbol{A^1}$ are spatial components (corresponding to x, y, and z coordinates) This was the 'Contravariant Form' (Uppercase indices) of a Four-Vector.

The 'Covariant Form' (Lowercase indices) of this Four-Vector looks like:

$$A_{\mu} = (A_0, A_1, A_2, A_3)$$

The covariant form's components are related to the contravariant components by a sign change in the time of spatial parts, respectively, using a Minkowski Metric tensor:

$$A^{\mu} = n_{\mu v} \cdot A^{v}$$

For "mostly-minus", it is: $\boldsymbol{A_{\mu} = (A^0, -A^1, -A^2, -A^3)}$
For "mostly-plus", it is: $\boldsymbol{A_{\mu} = (-A^0, A^1, A^2, A^3)}$

Four-Vectors are used to Transform properly under Lorentz transformations, making calculations frame-independent (Invariant).

The simplest and most fundamental example of a Four-Vector is:

$$X^{\mu} = (ct, x, z, z)$$

where:

01 – $\boldsymbol{X^0 = ct}$: Is the first component, hence of time scaled by $\boldsymbol{c}$ to result a length unit.

02 – $\boldsymbol{X^1 = x, X^2 = y, X^3 = z}$: Are the standard spatial coordinate s.

A compact notation of this, "Position Four-Vector" is:

$$x^{\mu} = (ct, \vec{x})$$

Where $\boldsymbol{\vec{x}}$ is an ordinary 3D position Vectors.

Another example of this can be a 'Four-Velocity', which is defined as the rate of change of its position (Position Four-Vector) with respect to its 'Proper Time'. The key idea of this example is that everyone is moving through the spacetime at the speed of light. Fir example, if we sit (stationary), we redirect 100% of this speed to time! But, if we move, we use some of this speed for both space and time. Meaning that the Total of moving through space and time is always the same, $\boldsymbol{c}$!

If you try to connect, this is why astronauts are thought to be aging slowly (Biologically). While this is true, it also remains a s a misconception because the harsh environments they go through, accelerates their biological aging. Thus, astronauts maybe be hardly younger than a blink of an eye.

A Four-Velocity can be written as:

$$u^{\mu} = \frac{dx^{m}u}{d\tau}$$

where:

01 – $\boldsymbol{u^{\mu}}$: Is the Position Four-Vector.

02 – $\boldsymbol{\tau}$: Is Proper Time, that one experiences, personally.

03 – $\boldsymbol{\frac{dx^{\mu}}{d\tau}}$: How your space time position changes per unit proper time.

Before we continue to learn to the components of the $\boldsymbol{u^{\mu}}$, we should first uncover how to achieve the proper time. The formula for calculating a Proper Time is:

$$\tau = \int d_{\tau}$$

In the formula above, we integrate or you can say add, infinitesimally small bits of proper time, altogether to get the actual Proper Time. And, to get this bit of Proper Time, written as $\boldsymbol{d_{\tau}}$, we will use this formula:

$$d_{\tau} = \sqrt{-\frac{ds^2}{c^2}}$$

where:

01 – $\boldsymbol{ds^2}$: Is the Spacetime Interval, used here to add the invariance to the Proper Time.

02 – $\boldsymbol{c^2}$: Is the Speed of Light squared

The reason we have negative sign before our spacetime interval is because the result of our spacetime interval is already a negative value, and obviously a negative value on a square root is senseless and not possible. Thus, we cancel that negative with another negative sign.

Now, we calculate the components of $\boldsymbol{u^{\mu}}$:

Time:

- $\boldsymbol{u^0 = \frac{d(ct)}{d\tau}}$ and because $\boldsymbol{\gamma = \frac{t}{\tau}}$

$\implies \boldsymbol{u^0 = c \times \frac{dt}{d\tau}} \implies \boldsymbol{u^0 = c \times \gamma}$

$\implies \boxed{\mathbf{u^0} = \boldsymbol{\gamma c}}$

Spatial:

- $\boldsymbol{u^1 = \frac{dx}{d\tau}, u^2 = \frac{dy}{d\tau}, u^3 = \frac{dz}{d\tau}}$

$\implies \boxed{\mathbf{u^1} = \boldsymbol{\gamma v_x}}, \boxed{\mathbf{u^2} = \boldsymbol{\gamma v_y}}, \boxed{\mathbf{u^3} = \boldsymbol{\gamma v_z}}$

We know that $\boldsymbol{\gamma}$ is the Lorentz Factor, which is equal to the coordinate time (Actual Time) divided by the Proper Time, because for example moving , slows the proper time than the normal time. And because of this Lorentz factor formula, we get our first component as $\boldsymbol{\gamma c}$ And of the spatial ones, we just multiply the simple derivative of that spatial coordinate and time with the Lorentz Factor.

Here are all the components shown:

$$u^{\mu} = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)$$

Another example of this Four-Vector can be a “Four-Momentum”. Four-Momentum is simply mass times Four-Velocity:

$$p^\mu = mu^\mu$$

Where $\boldsymbol{m}$ is a ‘Rest Mass’, which is a mass measured of an object when it was at rest.

And since $\boldsymbol{u^\mu = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)}$

$$p^\mu = (m\gamma c, m\gamma v_x, m\gamma v_y, m\gamma v_z)$$

There is something special in this time component $\boldsymbol{m\gamma c}$ of $\boldsymbol{p^\mu}$, let us see what it is:

$$p^0 = m\gamma c \implies \frac{mc}{\sqrt{1-\frac{v^2}{c^2}}}$$

This $\boldsymbol{p^0}$, if we multiply it $\boldsymbol{c}$, we get the total energy of a particle in Special Relativity!

$$E = p^0 \cdot c = m\gamma c \implies m\gamma c^2$$

Because if this Energy and time component of $\boldsymbol{p^\mu}$ relation, we can also write the time component of $\boldsymbol{p^\mu}$ as:

$$\boldsymbol{p^0 = \frac{E}{c}}$$

Spatial components of the Four-Momentum also matters, they can be written as:

$$p^1 = m\gamma v_x, p^2 = m\gamma v_y, p^3 = m\gamma v_z$$
$$\text{OR}$$
$$\vec{p} = m\gamma\vec{v}$$

If a particle moves quite slowly $\boldsymbol{(v < c, \gamma \approx 1)}$:

$$\vec{p} \approx m\vec{v}$$

This is the classical momentum formula! But, if a particle moves faster (close to $\boldsymbol{c, \gamma > 1}$):

$$\vec{p} = m\gamma\vec{v}$$

Hence, moving faster results in a greater momentum. Let us take an example:

- A particle has a mass $\boldsymbol{m = 1}$kg and a velocity $\boldsymbol{v = 0.8c}$ in X direction

$$\gamma = \frac{1}{\sqrt{1 - \frac{(0.8(3\times10^8))^2}{(3\times10^8)^2}}} = 1.67$$

- If we just use classical formula of momentum:

$$\boldsymbol{p = mv \implies 1 \times 0.8(3 \times 10^8) = 0.8c}$$

- But, if we introduce our Lorentz Factor:

$$\boldsymbol{p^x = m\gamma c}$$

$$\boldsymbol{\implies 1 \times 1.67 \times 0.8(3 \times 10^8) = 4.008 \times 10^8} \text{ or } \boldsymbol{1.33c}!$$

Hence, relativistic momentum is bigger! This explains that the closer a particle is moving to the speed of light, the rapidly grows the Lorentz Factor, and momentum too. Because of this a lot of force is required to just increase p to increase the velocity slightly. And thus, it is impossible of any particle with mass to reach speed of light, Only massless objects like 'Photons' (light particles) can travel at light speeds.

There is a relation between Energy and Momentum represented, or better expressed as:

$$E^2 = (pc)^2 + (mc^2)^2)$$

Which means 'Total energy squared is equal to whole square momentum energy, summed with the whole square of rest energy. Here:

01 – $\boldsymbol{E = m\gamma c^2}$: Is all the energy of a particle (Rest plus/and Kinetic).

02 – $\boldsymbol{pc = m\gamma vc}$: Is just momentum multiplied by $\boldsymbol{c}$ to have energy units.

03 – $\boldsymbol{m^2}$: Is Is the 'Rest Energy', which is locked in the mass of an object itself, even when its stationary.

This formula altogether is just like the old famous Pythagorean Theorem! $\boldsymbol{(a^2 = b^2 + c^2)}$

▮ Christoffel Symbols ▮

Christoffel symbols describe how space is curved, by telling use how basis vectors change as we move around.

To understand the need of Christoffel Symbols, let us use two different analogies:

- Imagine you're on a flat desert. You have a compass. You are at Point 'A' and You point your arrow at North. Then, you start walking toward another point 'B'.
- You look at your arrow, and it still points exactly at North, even after you moved.
- The Grid (North/South/East/West) never changes!

In this flat world, Christoffel Symbols = 0. There is no need for 'correction factors'.

Now, let us take another, Sphere analogy:

- Imagine you're on a Planet, at the Equator (an imaginary division line that perfectly cuts halfway between North and South Poles)
- You point an arrow towards, let's say North again, and start walking along the equator.
- Look at your arrow, it is rotating as you move, because the Planet you are in is Sphere! This is the same as what we learned before, that the Light itself moves in a straight line, but curves due to the curvature of the Space. Therefore, the arrow in this example bent due to the curved planet you are walking on.

This is where Christoffel Symbols shine! Because moving from Point A to B on a curved ground curves or better, 'tilts' the grid lines as well. So, if you want that your arrow points exactly at one direction NO MATTER WHAT, then use Christoffel Symbols.

Christoffel Symbols calculates how much you need to tilt your arrow (Vector) to cancel the automatic tilt the curve surface gives, and so that the arrow (Vector) points in the same place, or better stays 'straight'.

The formula of Christoffel Symbols is:

$$\Gamma^k_{ij} = \frac{1}{2} \cdot g^{kl} \left(\frac{\partial g_{lj}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

Before we dive into the formula more, we must understand the indices:

01 – $\boldsymbol{i}$: Is the Direction you're are Walking.

02 – $\boldsymbol{j}$: Is the Direction your arrow is pointing at.

03 – $\boldsymbol{k}$: Is the direction in which your arrow tilts naturally.

The $\boldsymbol{l}$ index is a summation index. It loops through each dimension, as rotation can occur in each dimension, and add them all up. We can call it the 'Searcher' Index.

$$\Gamma^k_{ij} = \frac{1}{2} \cdot g^{kl} \left(\frac{\partial g_{li}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

where:

01 – $\boldsymbol{\Gamma^k_{ij}}$: If walking in direction $\boldsymbol{i}$, with an arrow pointed at direction $\boldsymbol{j}$. How much does the arrow naturally tilt in the direction $\boldsymbol{k}$?

02 – $\boldsymbol{\frac{\partial g_{li}}{\partial x^i}}$: Measures how much the grid lines bend because you are moving in the direction $\boldsymbol{i}$.

03 – $\boldsymbol{\frac{\partial g_{il}}{\partial x^j}}$: Measures how much the grid lines bend in the direction of your arrow, $\boldsymbol{j}$.

04 – $-\boldsymbol{\frac{\partial g_{ij}}{\partial x^l}}$: Measures only the stretch of the two direction of your movement & arrow's point, $\boldsymbol{i}$ and $\boldsymbol{j}$ respectively. We subtract this derivative because the first two derivatives calculate the stretch with the bend. We don't need that bend, as it is of the Coordinate system which we don't care about, hence we discard/cancel it through this derivative.

05 – $\boldsymbol{g^{kl}}$: Is the ‘Inverse Metric’, used as a translator. If the grid lines on the ground are very ”tight” (bunched together), a small bend in the grid might cause a huge tilt in your arrow. If the grid lines are very ”loose” (spread far apart), that same bend might barely move your arrow at all. This translator here takes everything we calculated inside the parenthesis and translates it into the actual tilt of the arrow, based on how wide or narrow the grid is at that point.

06 – $\mathbf{\frac{1}{2}}$: Measuring distances and angles on a grid, we always do it using squares, like the pythagorean theorem. So, this fraction just cancels out the square, so the final answer isn’t about the arrow tilt twice.

▮ Partial Derivative of Metric Tensor ▮

Because the numbers in a Metric Tensor tell us how to measure distances at a point, in curved spaces these numbers change as we move. Hence comes the Partial derivatives, which are used to measure how these metric values changes as we move through a dimension/direction.

We can see these derivatives as either:

$$\frac{\partial g_{\mu v}}{\partial x^{\sigma}}$$

OR

$$g_{\mu v,\sigma}$$

where:

01 – $\boldsymbol{g}$: Is the metric tensor

02 – $\boldsymbol{\mu}$: Is a row of the metric tensor

03– $\boldsymbol{v}$: Is a column of the metric tensor

04 – $\boldsymbol{x^{\sigma}}$: Is the coordinate direction, e,g 1 for x, 2 for y, etc.

05 – Comma$(\boldsymbol{,})$: Is the Physics Short -hand for "Take the derivative, with respect to" what ever is after the comma.

Lets do a walkthrough, shall we? We will use 2D Polar Coordinates.

Let us first map each coordinate to the sigma index on top of x, and revise the Polar Metric:

$$x^1 = r \qquad\qquad \boldsymbol{g_{\mu\nu}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$
$$x^2 = \theta$$

Now, we use the Partial Derivative to measure the change of all four values in our Polar Metric, for each coordinate direction, hence 8 possibilities. First up is the coordinate direction 1 ($\boldsymbol{r}$).

011 – $\boldsymbol{g_{11}(1)}$: Does 1 change when $\boldsymbol{r}$ changes? No, in partial derivatives of a constant value always result in a 0. Hence, the result is zero.

012 – $\boldsymbol{g_{12}(0)}$ and $\boldsymbol{g_{21}(0)}$: Does 0 change when $\boldsymbol{r}$ changes? No, same reason as before, zero result.

013 – $\boldsymbol{g_{22}(r^2)}$: Does $\boldsymbol{r^2}$ change when $\boldsymbol{r}$ changes? YES! Using very basic calculus power rule, the derivative of $\boldsymbol{r^2}$ will be $\boldsymbol{2r}$.

Now, we loop through the second dimension/coordinate direction ($\boldsymbol{\theta}$)

021 – $\boldsymbol{g_{11}(1)}$: Does 1 change when $\boldsymbol{\theta}$ changes? No, same zero result.

022 – $\boldsymbol{g_{12}(0)}$ and $\boldsymbol{g_{21}(0)}$: Does 0 change when $\boldsymbol{\theta}$ changes? No, zero result.

023 – $\boldsymbol{g_{22}(r^2)}$: Does $\boldsymbol{r^2}$ change when $\boldsymbol{\theta}$ changes? No! Because if the coordinate letter you are changing ($\boldsymbol{\theta}$) doesn't match the other letter we are checking if it changes or no ($\boldsymbol{r^2}$), it drops a zero.

These are the same derivatives as we learned, in Christoffel Symbol's Formula. We just got one derivative out of the 8 possibilities, which is $\boldsymbol{g_{22}}$ for the first dimension, with the value of $\boldsymbol{2r}$.

▮ [Demystification] ▮

Before we continue ahead, this is an optional chapter, made to clutter up the difference between Metric Tensors, their partial derivatives and Christoffel Symbols, as they might all seem to do one thing for most of you.

Starting with Metric tensor. Metric tensors are values that tell us a factor which we can use **at each step**, to measure distances in a curved space. Fixes the pythagorean theorem, basically.

But, so these Metric Tensors can gives correctional factors **at each step**, we take their Partial Derivative. So, that we know that moving in what direction, results in what. It's like a predictor almost.

Whereas a Christoffel Symbol is used to take those directional changes (Metric Derivatives) and use them to calculate how much a Vector must bend to stay straight! So, it doesn't rotate unnecessarily with the curved space.

▮ Calculation using Christoffel Symbols ▮

Let's use our 2D Polar grid example ($\boldsymbol{x^1 = r, x^2 = \theta}$) to find the Christoffel Symbol. We will find the Christoffel Symbol for only $\boldsymbol{g_{22}}$ of first dimension, as it was the only non-zero derivative we found.

Decide a partial derivative within the parentheses of the Christoffel Symbols, make it's values accordingly:

$$\frac{\partial g_{ij}}{\partial x^l} \implies \frac{\partial g_{22}}{\partial x^l} \implies i = 2, j = 2, k = 1$$

$\boldsymbol{k = 1}$ so that the calculation stays alive and $\boldsymbol{g^{kl}}$ doesn't result zero.

Now, we plug in the values:

$$\Gamma^1_{22} = \frac{1}{2} g^{1l} \left(\frac{\partial g_{l2}}{\partial x^2} + \frac{\partial g_{21}}{\partial x^1} - \frac{\partial g_{22}}{\partial x^l} \right)$$

As $\boldsymbol{l}$ is a summation index with two option ($\boldsymbol{r}$) or ($\boldsymbol{\theta}$), it goes for both. But, for the second dimension, it results in zero, so we skip and talk only for the first dimension.

$$\frac{1}{2} g^{11} \left(\frac{\partial g_{12}}{\partial x^2} + \frac{\partial g_{21}}{\partial x^1} - \frac{\partial g_{22}}{\partial x^2} \right)$$

The first two terms will result in zero, only the last will result in $\boldsymbol{2r}$, as we did before.

$$\Gamma^1_{22} = \frac{1}{2} g^{11} \left(0 + 0 - 2r \right)$$

$$\implies \Gamma^1_{22} = \frac{1}{2}(1) \left(-2r \right)$$

$$\implies \Gamma^1_{22} = -r$$

Hence, the Christoffel symbol for $\boldsymbol{r^2}$ changes when $\boldsymbol{r}$ does, will be $\boldsymbol{-r}$. The minus sign means that your arrow is twisting towards the center of the circle. Meaning, we must pivot our arrow inward by $\boldsymbol{r}$ to keep it straight.

A quick secret before we move forward! In the world of General Relativity, Christoffel Symbols actually come in two flavors: the *First Kind* and the *Second Kind*. The formula and calculations we just went through are specifically for the Second Kind.

Here is the simplified way to tell them apart and understand what they do:

First kind

$$\Gamma_{kij} = \frac{1}{2}\left(\frac{\partial g_{ki}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k}\right)$$

Notice: They have all lower indices $\boldsymbol{\Gamma_{kij}}$. Also, there is no upper index floating on top! Basically, they measure the raw, unadjusted bend of your grid lines. Because they don't use an inverse metric tensor $\boldsymbol{g^{kl}}$ outside the parentheses, the answer you get is still stuck in the "raw units" of your local space.

Second kind

$$\Gamma^k_{ij} = \frac{1}{2} \cdot g^{kl}\left(\frac{\partial g_{lj}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l}\right)$$

Notice: They have one upper index and two lower indices $\boldsymbol{\Gamma^k_{ij}}$ because they use the inverse metric translator $\boldsymbol{g^{kl}}$ to "pull" one index up to the top! Basically, it takes that raw bend from the First Kind and translates it into a real, physical "tilt factor." It tells you exactly how many degrees or meters your vector arrow physically tilts in the direction of the upper index $\boldsymbol{k}$.

So the takeaway is, that the *First Kind* calculates the raw grid distortion. The *Second Kind* fixes it with a translator $\boldsymbol{g^{kl}}$ to tell you how your actual arrow bends. Because the Second Kind gives us the direct physical tilt, it is the one Einstein used the most to build his field equations!

The Covariant Derivative is the Ultimate tool in Differential Geometry for taking the derivative of curved surfaces.

In a complete flat space using standard grid lines, a regular partial derivative works perfectly. However, on a curved surface the grid lines themselves twist and bend. If you try to use a normal derivative there, your calculation will be wrong as it won't be able to differentiate between a vector that is actually changing and a grid that is bending.

The Covariant derivative fixes this. It calculates the raw changes of a vector and then uses Christoffel symbols to automatically subtract all the fake changes caused by the curving of the coordinate system.

The formula of Covariant Derivative is:

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda$$

where:

01 – $\boldsymbol{\nabla_\mu}$: Covariant Derivative Operator. The lower index ($\boldsymbol{\mu}$) specifies the coordinate direction we are walking in while measuring the vector's change.

02 – $\boldsymbol{V^\nu}$: The Vector we are analyzing. The upper index($\boldsymbol{\nu}$) specifies a specific component of the vector we are tracking (e.g., North component, South component, etc).

03 – $\boldsymbol{\partial_\mu V^\nu}$: Standard calculus derivative. Measures the raw change in the Vector's components on your coordinate system. Can't work for curved spaces.

04 – $\boldsymbol{\Gamma^\nu_{\mu\lambda}}$: The Christoffel symbol (of the second kind) that calculates how much the grid lines are bending at a point. Notice we said earlier to subtract all the fake changes that Christoffel symbols do, but then why are we adding it instead of subtracting. Well, it turns out that Christoffel symbol results in grid changes but those aren't fake, so the fake of those will be negative of these changes, hence $\boldsymbol{-\Gamma}$! And because we need to subtract the Christoffel symbols from the normal derivatives, we will add it, because remember the basic math rule: Minus and Plus is Minus! Hence, even using a plus sign, we still are subtracting!

05 – $\boldsymbol{V^\lambda}$: A Correctional Multiplier. It is the vector's component themselves. Fake changes of the grid bending depend directly on the size of the Vector (no. of components in a vector).

Notice that the dummy index ($\boldsymbol{\lambda}$) appears twice in the formula. Well, according to Einstein Notation, this index automatically triggers a loop that sums over every dimension in your space. For example, in a 4D spacetime, this index expands into four pieces: ($\boldsymbol{\lambda = 0, 1, 2, 3}$); thus gathering the grid corrections from every dimension.

The need of this is so that we can use derivatives but on curved surfaces as well. For example, a weather vane on a trampoline is facing North. And because the trampoline is curved, we can't just use normal derivatives to calculate how fast our vane is turning. Because normal derivatives are blind for curved surfaces. So, we instead take Christoffel symbol to know exactly how much the grid is bending, and just subtract it from a normal derivative.

We can also use the same formula on flat surfaces. Because, the grid lines won't be bending there, the Christoffel symbol will naturally just be 0!

Before we move on forward, there is one last piece of mathematical cleanup we need to address. The formula we just learned: $\boldsymbol{\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda}$ is specifically for a standard vector (upper index). But what happens if you want to take this derivative of a co-vector ($\boldsymbol{V_\nu}$), where the index is on the bottom?

Well, there is just a small concept change we need to address:

$$\nabla_\mu V_\nu = \partial_\mu V_\nu - \Gamma^\lambda_{\mu\nu} V_\lambda$$

Because Co-vectors transform inversely to standard vectors we use a minus sign before the Christoffel symbol, because the grid correction factor must be subtracted instead of added to correctly cancel out the fake grid distortion.

The main take-away is that Upper index uses a plus, Lower index uses a minus before the Christoffel Symbol! Nothing more to it.

NOTE: These were Greek Indices. But, because we have mostly gone through Latin Indices, you can easily replace: ('$\boldsymbol{\mu}$' by $\boldsymbol{i}$), ('$\boldsymbol{\nu}$' by $\boldsymbol{j}$), and('$\boldsymbol{\lambda}$' by $\boldsymbol{l}$).

Parallel Transport is walking across a curved grid with a vector, but ensuring that the direction that the vector is facing at doesn't change.

In flat spacetime, keeping a vector straight is simple enough: you simply leave its components completely alone. However, on a curved manifold, grid lines bend and twist beneath your feet. If you keep the raw component values constant, the vector will rotate relative to the actual geometry of the space. To prevent this fake rotation, you must continuously apply correction factors to the vector's components as you move. This deliberate adjustment of using correction factors is what comes under a Parallel Transport. Also, mathematically we say a vector is parallel transported if its 'covariant derivative' along the path is exactly zero.

The Mathematical representation of Parallel Transport is:

$$\frac{dV^\nu}{d\lambda} = -\Gamma^\nu_{\mu\alpha} T^\mu V^\alpha$$

where:

01 – $\boldsymbol{V\nu}$: The specific vector component we are actively updating.

02 – $\boldsymbol{\lambda}$: This is basically a milestone tracker. It runs like a stop-watch and marks milestones (journey's start, half-way mark, end, etc.) so that we can know the vector's position at every point.

03 – $\boldsymbol{\frac{dV^\nu}{d\lambda}}$: The raw vector change we must apply to our vector's coordinate component to parallel transport the vector. Or in simple words: "How much does my coordinate component change at the rate of the 'milestone tracker', completely independent of how weirdly stretched the grid lines are."

04 – $\boldsymbol{-\Gamma^\nu_{\mu\alpha}}$: Christoffel Symbol of the second kind, that calculates how much the grid lines are bending, so that Parallel Transport Equation can use those calculations to figure out the exact physical vector updates needed. Notice: we use a minus sign here and that is because the bending of the grid lines naturally adds a fake $+5$ to our vector components (for example), so we must actively subtract that -5 to keep the vector physically straight.

05 – $\boldsymbol{T^\mu}$: Our velocity vector as you travel along a path.

06 – $\boldsymbol{V^\alpha}$: $\boldsymbol{\alpha}$ is the "Searcher" dummy index om the main Vector, similar to the dummy index from the previous chapter. It sums over every single dimension in your space to gather the total distortion.

This book is still in progress. If you spot any type of mistake or an error, kindly open an issue immediately on the github "Repository".